Are speech recognisers still 98% accurate, or has the time come to repeal 'Hyde's law'?

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In 1969 S R Hyde of the Speech Research Unit in the UK proposed a law which stated: 'The accuracy of speech recognisers is 98%. It was quickly followed by its first corollary: 'Because the accuracy of speech recognisers is 98%, tests must be arranged to prove it'. Inspired by 'Hyde's law', this paper reviews speech recogniser evaluation methods which have been used in the past, examines more closely the principles behind them and then goes on to show an alternative means for testing speech recognisers and systems. Finally, the future role of Hyde's law is questioned.

1. Introduction

In BT, automatic speech recognition (ASR) has already been applied to hands-free car phones, repertory dialling in networks and the control of answering services, and is now entering large-scale applications such as call centres and directory enquiries (see Chapter 1).

World-wide, over 70 companies [1] have speech recognition boards, algorithms or systems among their products and there are several hundred research institutes and laboratories actively developing advanced systems. Among these are the major telecommunications providers, for whom ASR is an essential en-abling technology, providing customers with direct access to machine-based information — simply by picking up the phone.

BT is especially active in ASR (see Chapter 12) [2], but as a **user** as well as a developer. As such, BT needs to know not just what is available, but what is the very best. Only by searching out and deploying the 'best of breed' can customers be given the best possible service. BT is not alone in this, and in recent years much attention has been directed towards developing test methods. The US-based Advanced Research Projects Agency (ARPA) organisation (whose motto is: 'Progress through evaluation') regularly organises competitions to compare the various technical solutions developed within that programme. Within Europe the COST project 'Speech Technology Assessment', and two ESPRIT projects 'Speech Assessment Methodology' and 'Speech Quality Assessment in Language Engineering' [3] have provided a European focus for developing assessment methods. Most of these programmes have a broadly based remit and target both the large-vocabulary speech understanding systems and telephony-based applications. A major interest therefore is in evaluating speech recogniser performanc in telephone networks and it is that which forms the basis of this chapter.

1.1 A change in emphasis

Only ten years ago commercial speech recogniser evaluation was dominated by issues such as power consumption, manufacturing cost, chip count and speed of response. At that time it was a major challenge to squeeze algorithms into the available processors and development effort was often directed at increasing code efficiency and 'cutting corners'.

The processing capacities of workstations have currently reached a stage where many speech recognition algorithms run in real time (see Chapter 1). As a consequence, two major new markets, one based upon telephony and one upon the 'voice activated typewriter', have emerged. Although both rely upon similar core technology (usually based upon hidden Markov models (see Chapter 8)), their realisations are very different. Contrasts are summarised in Table 1.

Table 1 Contrasts between telephony and office-based systems.

	Telephony	*Workstation/voice typewriter*
Vocabulary size	Usually 2—20 words occasionally up to 2000	5000 to 60 000 words
Flexibility in vocabulary	Preferably fixed — any changes may compromise performance	High — new words can be added quickly and easily
Adaptability	Little opportunity in most simple applications for continuous adaptation to talkers	Supervised learning strategies allow continuous improvement with use
Speaker variability	Speakers highly variable; wide range of accents, levels and vocal characteristics; often naive users; untrained	Only one speaker to whose precise characteristics and mannerisms the recogniser can adapt; opportunities for training
Signal quality	Extremely variable due to handsets, background noise and transmission interactions	Consistent — usually deploys a high-quality noise-cancelling microphone in a stable quiet environment; no transmission distortion
Word error rate	MUST be low to avoid invoking excessive error-correcting dialogues.	Higher rates tolerable as screen-based correction is quick and easy.

1.2 Background to performance measurement

Since the first demonstrable speech recogniser was constructed by Davis et al [4], engineers and scientists have tried to find reliable and realistic ways of measuring the performance both of the devices themselves and the systems into which they are built.

Until speech recording was possible, the only way to measure was using ‘live’ speakers. This was the method employed by Davis et al [4], where an accuracy of 97% (3% error rate) was claimed on the digits in a number of ‘speaker-dependent’ tests. Perhaps this set the standard as, over the following decades, most publications quoted similar results.

These consistent and surprisingly good results did not go unnoticed by researchers and in 1969 S R Hyde, of the Joint Speech Research Unit in the UK, proposed his ‘law’ stating that: ‘The accuracy of speech recognisers is 98%’, to be quickly followed by its first corollary that: ‘Because speech recognisers have an accuracy of 98%, tests must be arranged to prove it’ [5].

Although clearly tongue in cheek, ‘Hyde’s law’ highlighted the fact that the results obtained from speech recognisers were so sensitive to test methods, that by judicious or nefarious use of test material almost any accuracy could be obtained for any recogniser! His figure was allegedly chosen because that was what his customers always said they required.

However, there was no deceit involved. 98% accuracy was genuinely being achieved in the laboratories and was even being publicly demonstrated [6] at the time. The reasons for these high levels of accuracy was due to the quality of the speakers. Sometimes they were specially trained standard speakers but often they were the researchers themselves. All spoke in a clear, well-articulated fashion, but the key to their performance was in the consistency with which the words were spoken. Provided that the speakers trained the system with speech patterns which they could reproduce later, even simple recognition strategies would work well. With such speech material (and highly motivated users who knew how to make the recognisers work) good performance could be virtually guaranteed.

This serves to highlight several key aspects of testing:

- any measured performance is the result of an interaction between the recogniser and the test material,
- high-quality or standardised speech is not appropriate for realistic testing,
- testing and training material must always be separated,
- speech recognisers must work despite the environment — not because of it.

1.3 Results from real networks

With telephony systems there are additional issues to be considered. As simple recognisers began to be used in trial telephony services, published accuracy figures deteriorated. In 1987 a paper from Nynex [7] published measurements made on several of the leading telephony-based commercial speech recognisers and reported that the best performance obtained on the digits was only about 80% — and this was after some 'bad' data had been removed. Within BT similar results were being obtained. Figure 1 shows those obtained in a number of measurements undertaken in 1990 and 1991 and illustrates two more important effects.

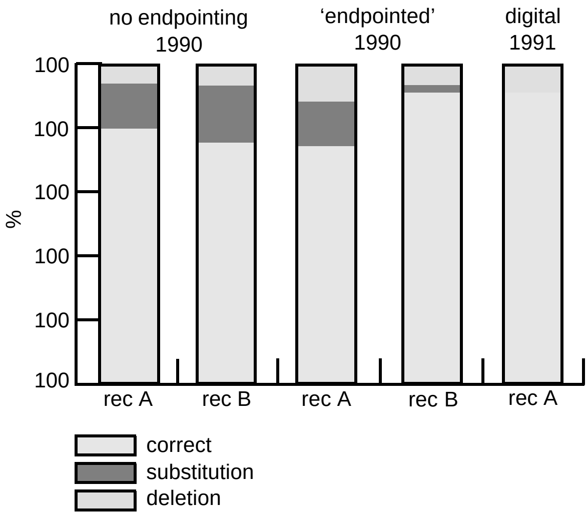

Fig 1 Effects of database 'treatment'.

Firstly, even though the data was collected using substantial speaker populations (in these experiments there were over 500 talkers per test) the measured performance changed significantly when this data was 'endpointed', i.e. where the speech signal was examined and cropped to remove the waste/silence before and after the word — this process normally makes it easier for the recogniser to 'catch' the word.

The accuracies obtained before and after endpointing are as shown in the first four columns of Fig 1. The results are unexpected, with one recogniser improving and the other deteriorating. This could possibly have been caused by this particular model trying to adapt to the now non-existent background noise.

The second effect was due to a change in the telephone network. The 1990 database had been collected over an analogue system (the 0800 overlay network), but the 1991 collection was over a fully digital network. At a stroke this single change in the BT network almost halved the measured error rate.

These examples serve to show that not only is there a strong interaction between the database content and the recognisers, but that it is sensitive to the collection processes and subsequent treatments.

Because of these strong interactions, for all practical purposes it is meaningless to refer to the 'accuracy' of a recogniser. This is why most experienced researchers only draw conclusions based upon 'head-to-head' comparisons and why, even then, results must be interpreted with caution.

1.4 Selection of speakers and speech material

At present, the only way to estimate the 'true' error rate of a telephony-based application is to create and run the service. In practice, logistic, ethical and legal reasons preclude this. Not only is testing normally required before a real service can be launched, but in a real service the time taken to collect sufficient data to estimate performance is often exceedingly long. It is also difficult to ensure sufficient coverage of those words which are crucial but appear only occasionally.

In practice, it is usually necessary to contrive a test database in the hope that it will be sufficiently close to the real application to allow comparisons to be drawn with regard to those factors (e.g. vocabulary type or recogniser type) which are important. This should contain data from customers prompted in a manner as close as possible to that of the genuine service. One way in which this may be done is illustrated in Fig 2.

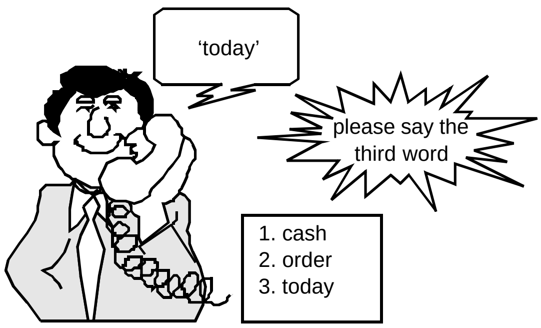

Fig 2 Indirect prompt method.

Prompted in this way, effects due to speakers 'echoing' accents or mannerisms in the voice used for prompting are minimised and elements of 'thinking time' and spontaneity are introduced. Nevertheless, no matter how much care is taken in collection (and sometimes because of the care taken), such a database can never be exactly the same as that of a real application and it will not predict the absolute performance. All that can be expected is that such a database will allow recognisers to be compared and give confidence that the recogniser which does best in the laboratory will also do best in the field. Provided that it spans the range of speech qualities and collection methods found in real applications and that good statistical methods are used to analyse and interpret the results, then it is possible to predict which recogniser will be best despite the vagaries of populations and collection methods.

How this analysis may be done is described later; before that, measurements must be made.

2. Characterising performance

2.1 Classifiers, detectors and errors

Once the database has been obtained and the words presented to the recognisers to be compared, the errors can be counted. There are several types of error which must be considered and the terminology used is often confusing, having been extensively borrowed from other disciplines, e.g. radar and filtering. It is not uncommon, for example, to see terms such as deletion and rejection used synonymously in the literature. Generally context and common sense must be used as a guide. The following provides a self-consistent terminology which will be adopted in the rest of this chapter.

- A speech recogniser can ‘get it right’ by:

  — correctly classifying all ‘valid’ input words — this is true acceptance;

  — correctly rejecting (ignoring) all ‘invalid’ inputs — this is true rejection.

  For example, the set of ‘valid’ inputs to a digit recogniser would be ‘zero..one.. two .. three... up to ...nine’. The set of ‘invalid’ inputs would be any other words or noise ....‘sixty’, ‘fence’, ‘cough’.....etc. These are also known as out-of-vocabulary (OOV) words when used for testing.

- Also a recogniser can ‘get it wrong’:

  — when a valid word is presented, but the match is not ‘good enough’ for the recogniser — this is a false rejection, or deletion error;

  — when a valid word is presented but the classifier gets it wrong — this a substitution error;

  — when an ‘invalid’ input is presented and the recogniser fails to reject it — this is an insertion error.

These last two types of error are also sometimes known as false acceptances.

Although the performance of the recogniser is determined by its error profile the best performance is not necessarily achieved when the error rate is a minimum, as the occurrence of a false acceptance has different consequences from a false rejection. Usually the cost of a false acceptance is considered to be higher than that of a false rejection, as not only is the latter detectable but is amenable to recovery strategies using error-correcting dialogues.

All these error types may be analysed by considering speech recognition as a filtering and detection process, as shown in Fig 3.

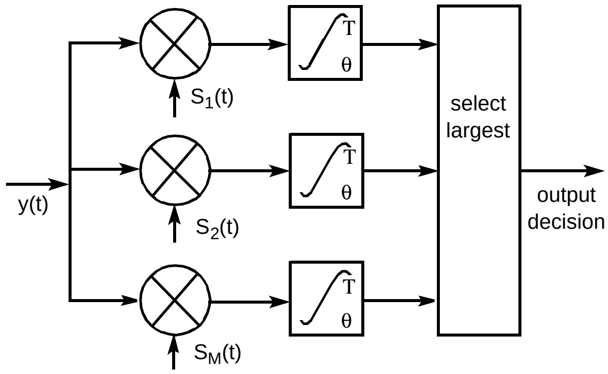

Fig 3 Classical M-ary filter and detect process.

At its simplest, a speech recogniser receives signals which it labels as belonging to one of M classes. Viewed in this way a recogniser is a classical ‘M-ary’ filter [8] with an upper theoretical performance limit set by the extent to which probability distributions of the candidate signals overlap. For example if the words ‘hut’ and ‘hat’ are used and the speakers are drawn from a wide range of English accents there is an irreducible confusability as ‘hut’ spoken by some southern English speakers sounds just like ‘hat’ spoken by some northern English speakers. The costs of such vocabulary-dependent recognition errors are not symmetric, and this must be taken into account in the application. As an example of this asymmetry, consider the cost of confusing ‘bombs away’ and ‘close bomb bay’!

2.2 Ways of representing performance of isolated words

Although most modern telephony-based speech recognisers are capable of connected-word operation, the formidable problems of error detection and correction means that this is not yet widely deployed in the simpler systems for public use [9]. Instead, such recognisers are constrained to operate in isolated-word mode with the 'connected' facilities used as a 'safety net', often wordspotting the occasional utterance where the talker fails to speak in an isolated fashion.

In principle, the process of measuring recogniser error rates of such systems seems straightforward. All that is necessary is to present the speech recogniser with isolated words spoken by number of speakers and count the errors. Such error counting forms the basis for most published results — often represented in terms of %correct or %accuracy

The usage of these terms (for isolated words) is discussed in Winski et al [10]. In this chapter the convention adopted is:

% correct = 100% − substitution errors − deletion errors
% accurate = 100% − substitution errors

The distinction is made in this way because %correct is the figure which the user of an application experiences, whereas %accuracy is the figure from 'forced choice' tests — where the recogniser always responds to an input, such as is usually the case in database testing.

Less often a separate measurement is made using an out-of-vocabulary test set and the two figures combined to give a single figure of merit [11].

This average error-rate forms a useful single index, but a richer way of presenting performance is provided by the confusion matrix. An example based on the four words 'north', 'south', 'east', and 'west' is shown in Fig 4. This shows the pattern which might emerge when 100 examples of each word are presented to a recogniser.

		word presented			
		north	south	east	west
word recognised	north	93	0	2	0
	south	4	98	0	0
	east	0	2	91	7
	west	3	0	7	93

Fig 4 Substitution error confusion matrix.

Figure 4 shows, for example, that the word 'east' was correctly recognised 91 times, but was misrecognised as 'north' twice and as 'west' on seven occasions.

The matrix highlights inconsistency in word performance but in this form only shows substitution errors.

Other error types can be incorporated by augmenting this confusion matrix with an extra row and column to include the false acceptances and deletions (see Fig 5). Here a separate out-of-vocabulary set of test words must be used.

The practical advantage of the augmented confusion matrix is that trade-offs between the error types are apparent. For example, if the recogniser 'acceptance' criterion is tightened up (all other conditions being the same), then the number of substitution errors reduces and the number of 'out-of-vocabulary' words now being correctly deleted increases. However, these gains are at the cost of reducing the number of good matches of valid words, as may be seen by comparing the diagonals of the matrices.

		word presented				
		north	south	east	west	OOV words
word recognised	north	68	1	0	1	<u>13</u>
	south	3	89	1	1	<u>12</u>
	east	1	0	85	2	<u>7</u>
	west	0	0	4	84	<u>8</u>
	deletion	<u>28</u>	<u>10</u>	<u>10</u>	<u>12</u>	<u>60</u>

(a) Results with a low threshold.

		word presented				
		north	south	east	west	OOV words
word recognised	north	54	0	0	0	<u>3</u>
	south	0	78	0	0	<u>0</u>
	east	0	0	75	0	<u>1</u>
	west	0	0	1	76	<u>2</u>
	deletion	<u>46</u>	<u>22</u>	<u>25</u>	<u>24</u>	<u>94</u>

(b) Results with a higher threshold.

Fig 5 Augmented confusion matrices — showing effects of different thresholds.

2.3 The impact of altering thresholds

Figure 6 illustrates what happens as the criterion for accepting or rejecting words is altered.

Here a ‘single word’ pattern-matching unit is shown responding to an ensemble of, firstly, the ‘correct’ valid words and, then, with ‘wrong’ out-of-vocabulary words. A low number (distance score) corresponds to a close match. As the speaker never repeats a word identically, utterances, even of the correct words, vary, some matching better than others. With enough samples, distributions of the ‘scores’ for both curves can be derived as shown in Fig 6.

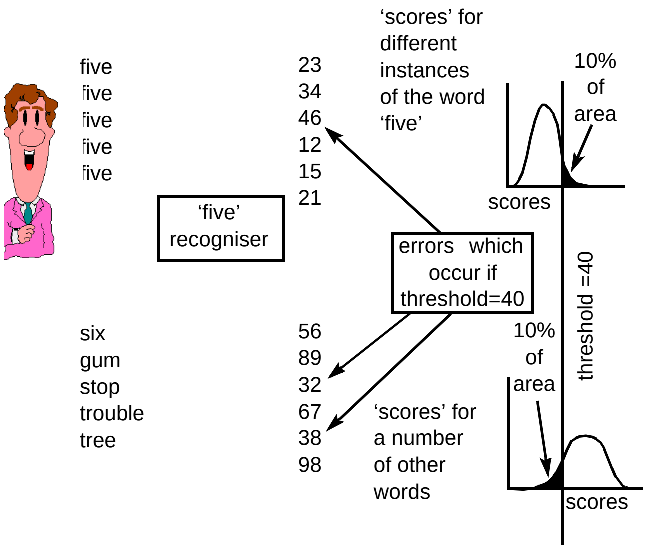

Fig 6 Pattern matcher responses.

The recogniser can only achieve perfect recognition if these two curves do not overlap. Unfortunately this is never the case in a real system and so, when a score is returned from a recogniser, a decision as to whether or not this is the valid word can only be made by comparing this score to a ‘threshold’. The threshold setting of 40 shown in Fig 6 is the point at which the number of the correct words rejected is equal to the number of invalid words accepted.

In Fig 6, about 10% of the valid words are shown to be falsely rejected and about 10% of the invalid words falsely accepted.

The effect of moving the threshold upon the error ratio is illustrated in Fig 7. This is analogous to the receiver operating curve (ROC) of filter theory. The sensitivity, which defines which curve is followed, is determined by the separation of the distributions.

For clarity, the above example was based upon a single-word recogniser. In a real recogniser the situation is complicated because every word undergoes multiple matching processes and all output scores compete with each other. In real systems, where each word recognition unit is

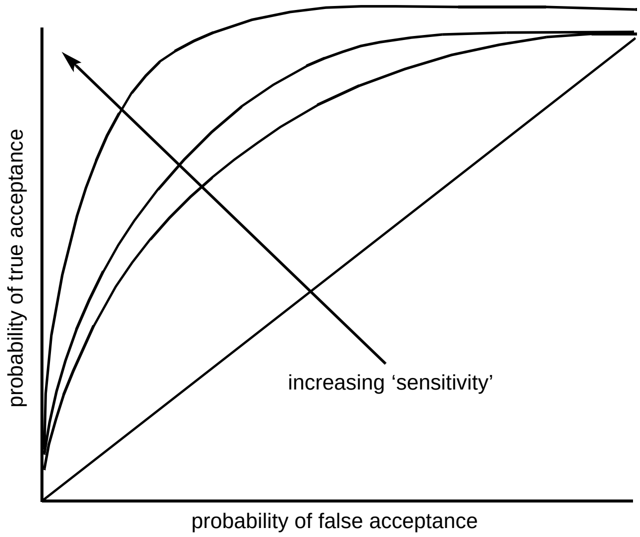

Fig 7 Receiver operating curve (showing effect of increased sensitivity).

independent, the means and variances of the scores from each class will differ as illustrated in Fig 8.

However, provided that the outputs of all the matching units have similar covariance distributions (or can be post-processed to give this characteristic) it is only necessary to select the best candidate and ensure that it is above the threshold. In practice, this threshold may be based upon a score from the ‘re-jection’ model. A comprehensive treatment of these issues is found in Devijver and Kittler [15].

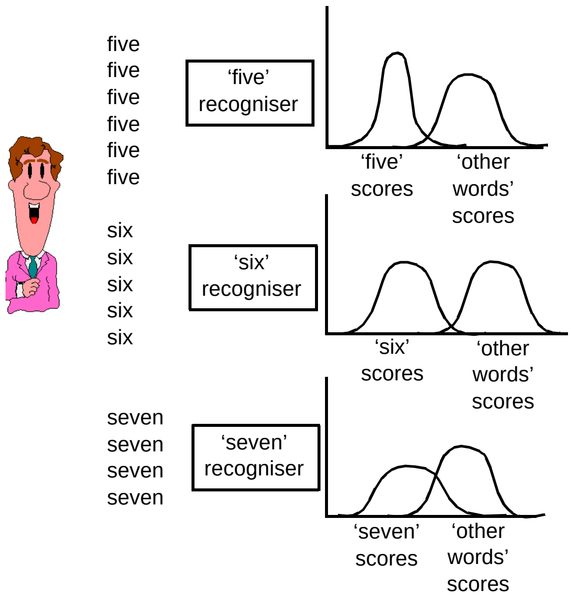

Fig 8 Normal multi-word recogniser.

2.4 Practical considerations for selecting the best threshold setting

If an application can be tailored so that it is guaranteed that speakers will only utter ‘valid’ words (a condition often assumed when the recognisers are tested on a fixed set of words), then the lowest overall error rate is achieved when the recogniser threshold is set to 'accept all'. However, the lowest is not necessarily the best and, as an undetected substitution error is usually more 'fatal' than a false rejection which is always detected, it is usual to err on the side of caution.

A ratio of at least 3:1 false rejections/substitution error-rate is normally desirable for typical tasks involving menu selections. This means that a speech recogniser which has an underlying error rate of (say) 4% will be operated at a point where the total error rate is doubled to 8%, of which 6% are false rejections and 2% substitution errors. Although the number of 'fatal' undetected errors is halved users will now experience twice as many total errors and will have to endure error-correcting dialogues on 6 out of 100 occasions.

3. Comparing the performance of real recognisers

3.1 An alternative strategy

So far it has been shown that any measured error rate is determined by:

- the 'true' error rate of the recogniser;
- the threshold setting strategy;
- the vocabulary;
- the 'out-of-vocabulary' test material;
- the population of speakers;
- the collection process.

The problem is that none of these can be meaningfully quantified.

One way to deal with such unquantifiable variability is to resort to larger and larger 'random' test databases in the hope that the larger the test, the better the coverage. This has been one of the trends in speech recogniser evaluation in recent years — spurred on by the need for equally large databases for training statistically based recognisers. However, this is only a valid strategy if the sample data is fully representative of the real application. Unfortunately, as has been shown, this is not the case.

This kind of difficulty is not unique to speech recogniser evaluation. The same problem arises in many other areas — especially those concerned with the life sciences where the test environments can never be specified nor repeated and are often highly variable. Many statistical techniques have been developed to deal with such instances, all based upon the original work of Fisher [13], and comprehensively discussed in Cochran and Cox [14].

3.2 Balanced databases

The first step involves designing the experiment so that the sources of variation are separated and samples are taken over a small number of fixed conditions. In this way, the material from each speaker is subjected to various treatments determined by the fixed conditions (see Fig 9).

Using these principles, a typical database consists of:

- 12 speakers — six male and six female;
- 10 words (digits) — repeated several times by each speaker for each condition;
- 3 telephone lines — short (500m), medium (1.5 km) long (7km);
- 3 telephones (two electret and one carbon);
- 2 background noise levels.

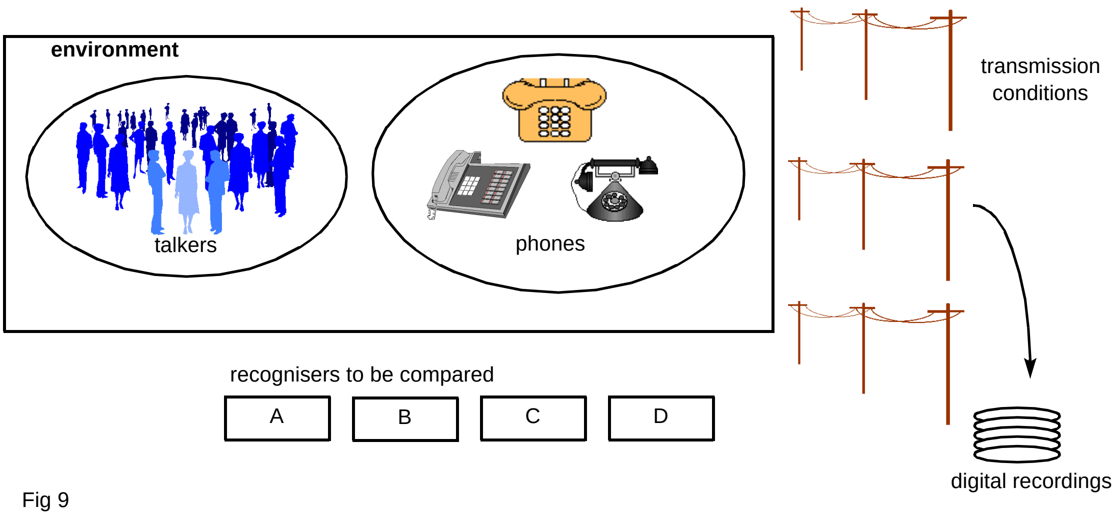

Fig 9 Principle of a balanced data collection.

Each speaker utters the same words into the three telephones over the three lines in two environmental noise conditions.

After the recorded material has been presented to the speech recognisers, the effects of the different treatments on error rates can be investigated. For example, as there are three sets of data containing speech from the same people speaking over each type of telephone line, the effects of the telephone lines can be compared. Similarly, as half the data is from males speaking over exactly the same wide range of conditions as the females, the results can be compared to see if there is a difference between male and female speech despite the variability of lines, phones, recognisers, etc. There are also three groups partitioned according to telephones and, as the speakers have all spoken at least twice over every condition, there are two sets of data which differ only in time of recording. Also, as the same data can be presented to different recognisers, it can be found not only if one is significantly better or worse than the others, but also over what range of conditions the best performance is obtained.

How this has been applied to speech recogniser testing is now illustrated with three specific examples.

3.3 *The fixed telephone network — are all digit recognisers the same?*

One such important question concerns the choice of recogniser technology or supplier. This is important for BT as, if there is no performance difference between the various recognisers, decisions can be made on the basis of other functionality or price. The question may be answered by restating the question as a null hypothesis that 'all commercial recognisers are the same' and the data analysed to see if this assertion is sustainable.

Consider what conclusions might be drawn from the results shown in Fig 10 where the data is partitioned into scores for male/female talkers, two different noise levels and a sequence effect.

The first observation is that background noise appears to be having a greater effect than the other two factors. Had the data not been derived from a balanced data set an appropriate test would have been a test of significance between the two noise conditions based on chi squared [15]. However, the balanced nature of the data means that better statistical tests may be made by comparing the various effects. For example, the male/female effects are so very much smaller than the noise effects that it would seem reasonable to use their small variability to 'benchmark' the relative importance of other respective factors. This basic principle underlies the technique of analysis of variance (ANOVA) which works by comparing variances determined in different ways — usually within and between the various classes, but often with a residual variance. This is the variance left after all those which can be accounted

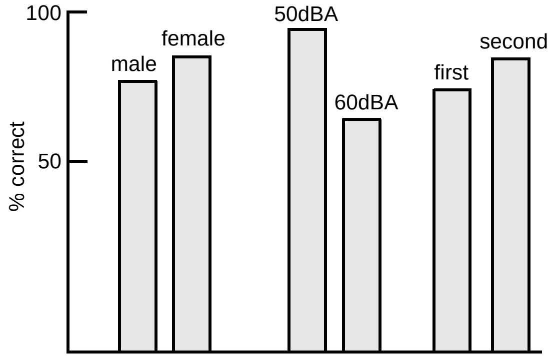

Fig 10 Male/female, background noise and sequence.

for are subtracted from the total, and is somewhat analogous to the background noise within the experiment.

This not only allows the significance of various factors to be estimated, but provides a means for estimating the overall reliability of the experimental process itself.

The partitioning of the data may be at many levels, and interactions between several factors may also be evaluated. For example, Fig 11 shows how individual talkers fare when tested over all lines, all phones, all words and all recognisers.

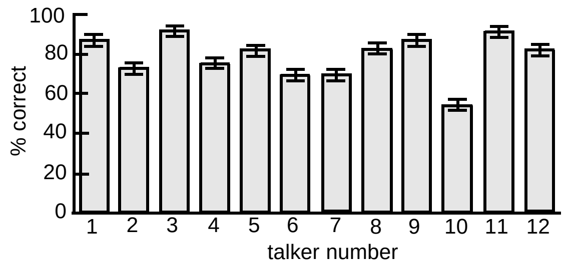

Fig 11 Recognition by talker.

Much more variability is apparent in Fig 11. This is not unexpected as there are now more degrees of freedom with more groups consisting of smaller numbers — which in itself will increase the between-group variation.

Figure 12 presents the same data — but this time showing performance by vocabulary item with each word averaged over all other factors.

Although the degrees of freedom are slightly different for these two cases (there are 12 talkers, but only 10 words), the variabilities are similar. Even greater detail can be found by examining the interactions between the words and the recogniser and Fig 13 suggests that one recogniser is not only better, but is much more consistent than others — a result which was borne out when the data was analysed using ANOVA.

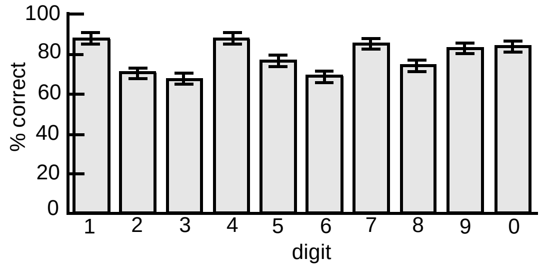

Fig 12 Effect of vocabulary item.

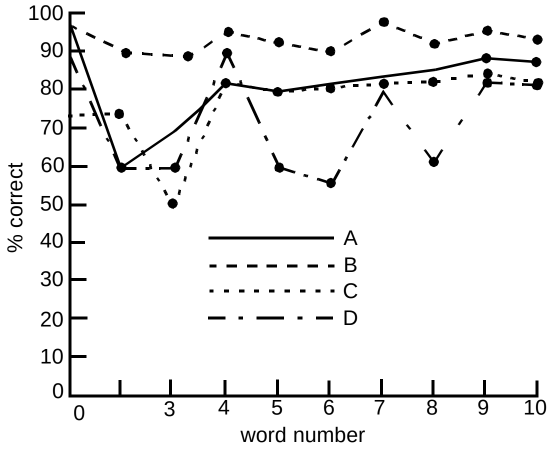

Fig 13 Word-recogniser interactions.

Although derived from a compact database, statistical analysis of the results shows that it is more than adequate to separate out major effects between the factors. It is clear, for example, that there is a significant difference between recognisers and that the best one is consistently better despite everything else.

One way in which all this data can be compactly represented is by plotting the proportion of variance attributable to each factor as shown in Fig 14. This shows that the single most important factor is the talker, followed by line, followed by recogniser. A more detailed description of these results can be found in Wyard et al [16].

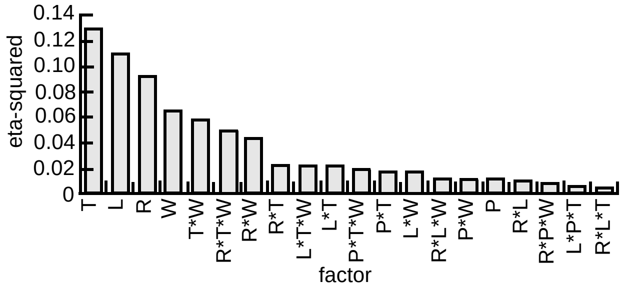

Fig 14 Proportion of variance (real lines).
(T : talker, W : word, R : recogniser, P : phone, N : noise, L : line)

In Fig 14 the eta-squared measure, η^2_X, is defined as:

$$\eta^2_X = \frac{SS_X}{SS_T}$$

where SS_X and SS_T are the effect and total sums of squares respectively [17].

The beauty of this method is that many different analyses are made on the same data. For example, Fig 15 shows the line-telephone interaction for each recogniser.

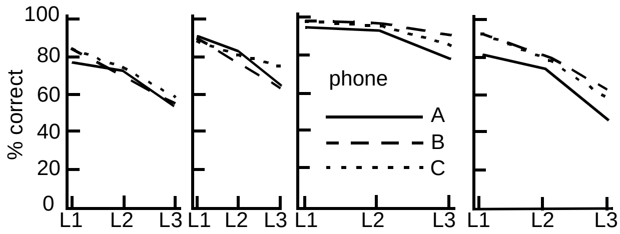

Fig 15 Line-telephone interaction for each recogniser.

3.4 *Do speech recognisers work with cell phones?*

The previous experiment examined the main factors affecting the performance of fixed telephones and showed that talker variability was the most important single effect, but what about cellular telephones? Not only are cellular telephone users much more likely to access services based upon speech recognition, but buttons on a cellular telephone are generally much less accessible than those on a fixed phone and makes speech recognition a very attractive option.

From a technical point of view, however, there are good reasons for suspecting that radio links might impair speech recognisers. In particular:

- fading,
- speech clipping due to voice switching,
- spectral distortion due to coding and bandwidth compression techniques,
- echo control techniques for delay,

might all be expected to compromise recogniser performance. The last two of these are incorporated in the GSM digital cellular network and are therefore of special interest for future networks; but first it is necessary to establish if all cellular telephones behave similarly.

3.5 *Cellular telephones — analogue system*

The design was similar to that used previously except that it now contained three cellular telephones. There was no ‘line’ effect. All were used at full signal levels.

As before, the data was collected from 12 speakers. Figure 16 shows the performance by talker.

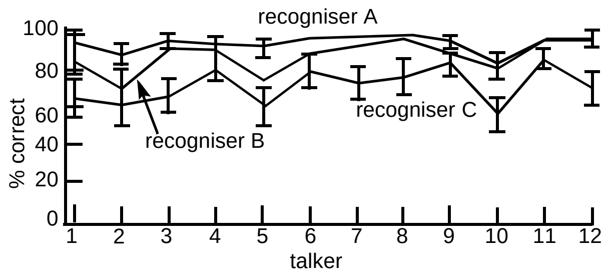

Fig 16 Mobile phone — percentage correct by talker.

This is a very similar pattern to that found in the fixed telephone experiments. The ranked variances for all factors obtained are given in Fig 17.

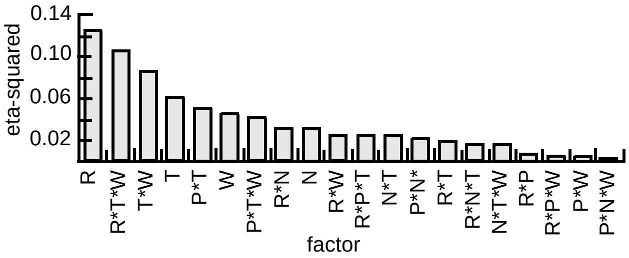

Fig 17 Ranked variances for all factors.
(T : talker, W : word, R : recogniser, P : phone, N : noise)

The overall conclusion is that users experience exactly the same performance from cellular telephones as they do over the fixed network — but only with certain recognisers. The recogniser performances were now more variable than the talkers, which was a different result to that found for fixed lines. They were also much more sensitive to background noise and in one instance there was a strong cellular telephone/recogniser/noise interaction. While a slight change in background noise level had little effect on the 'normal' phones, it had a much more marked effect on one of the recognisers when used with the cellular telephones.

3.6 GSM

The same principles were employed to compare GSM directly against the analogue phone set. The main results for most factors were found to be similar to those for the TACS system and the fixed network. Figure 18 shows talker variability for two recognisers.

Table 11.2 shows the results from a direct comparison between GSM and the fixed network.

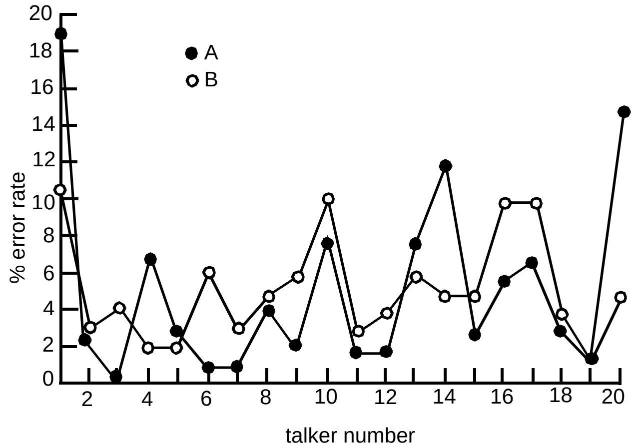

Fig 18 GSM — talker number for two recognisers.

Table 11.2 Direct comparison of GSM with the fixed network.

	B	*C*
GSM (average)	5.3%	4.25%
Fixed (average)	5.56%	6.11%

3.7 International access

Finally, the same technique was used to investigate how international telephony access affects overall performance. Already a substantial number of calls to UK information systems originate from other countries and trends in 'globalisation' mean that customers from anywhere in the world are now likely to be passed to a single call centre. If such call centres are to deploy speech recognition, they must be able to give comparable performance to speech originating from anywhere.

The data for this was gathered as part of the European Collaborative Project COST232 which collected a speech database from a range of participating European countries (see Fig 19).

In this case the balanced data consists of utterances of the (English) digits spoken by people in each of the participating countries. All speakers spoke over both fixed and radio networks and made separate, but identical, calls to both the UK and Italy.

The database contains 96 talkers spanning 12 accents and 24 transmission links. Half the speech is male and half is female. Every talker speaks the same words, over two phone types to two countries.

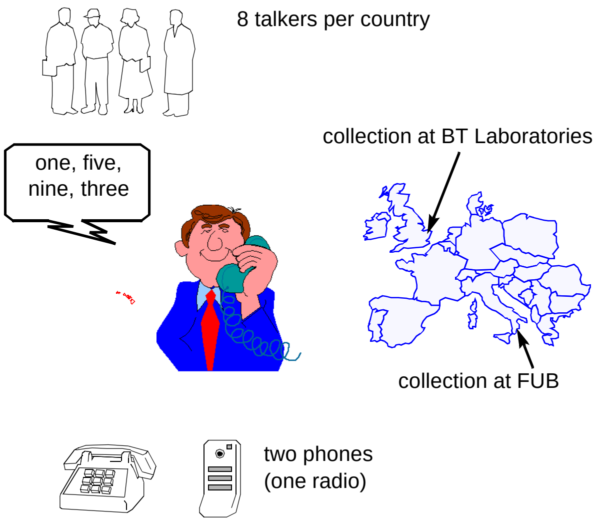

Fig 19 COST232 data collection process.

Figure 20 shows the general pattern of results obtained for the speech collected in the UK for three recognisers.

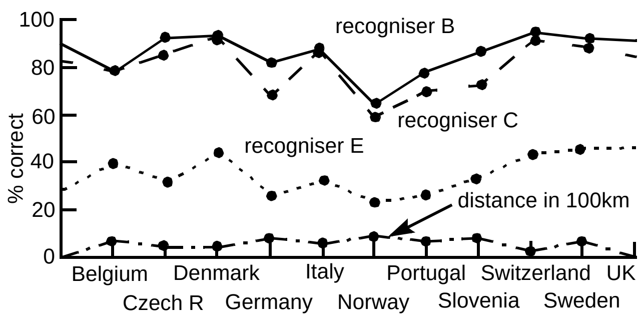

Fig 20 Performance by country (three recognisers) and distance.

As before, this design allows extensive comparisons to be drawn. Many of the results have confirmed those trends found within the UK. For example, the differences between male and female speech and between fixed and mobile were again found to be very small and the ranking of the recognisers was the same.

As would be expected the ‘local’ performance, with native English speakers talking over national networks, is high — but so are several others. Whether this is due to the ‘accent’ effect or the ‘lines’ effect is not yet known — once the full set of data is available this can be determined. The lower line in Fig .20, which shows the distance in 100s of kilometres from the UK, shows a trend which suggests the greater the distance, the worse the performance.

It is encouraging that a number of other ‘foreign’ countries appear to achieve a similar performance to that of the UK, suggesting that international boundaries need not be a major barrier to international service provision.

4. Conclusions

By undertaking experiments designed to compare the effects of specific factors, many of the problems traditionally associated with speech recogniser testing and evaluation can be avoided. Meaningful results can be obtained from small databases and the method also allows clear conclusions to be drawn from tests.

Some limitations, however, remain. In the main, the results presented here are those from one laboratory and are still limited in scope to the UK networks, the English language and a relatively small number of recognisers. To ensure that speech recognition continues to move forward other experiments must be done in other laboratories to replicate experiments, confirming or denying the generality of the methodology and the conclusions drawn.

Probably the most pressing need is for a suitable reference system to support the methodology so that experiments carried out in different laboratories and different languages can be compared. This could be based either upon human performance [18] or upon a software standard [19, 20]. Work towards such a reference has started, but, for such a reference to become an accepted standard, it needs to be tested and validated for several languages and environments.

But what of Hyde’s law? By formulating his law (and like many before, Kant [1], and later, Deming [21]), Hyde drew attention to the folly of trying to apply absolute measurements to that which cannot be meaningfully measured. Perhaps it is here that the true value of Hyde’s law lies. In the future, when methods based upon total quality replace those based upon ‘attainment of numerical goals’, Hyde’s law may be quietly repealed. Until then, it has a rightful place.

Acknowledgements

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[1] “It is a great and necessary proof of wisdom and sagacity to know what questions may be reasonbly asked. For if a question is absurd in itself and calls for an answer where there is no answer, it does not only throw disgrace on the questioner, but often tempts an incautious listener into absurd answers, thus presenting as the ancients said, the spectacle of one person milking a he-goat and of another holding the sieve”. ‘Critique of Pure reason’, Immanuel Kant (1724—1804).

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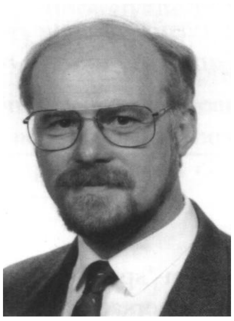

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